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# Master ARIMA: Your Guide to Time Series Forecasting

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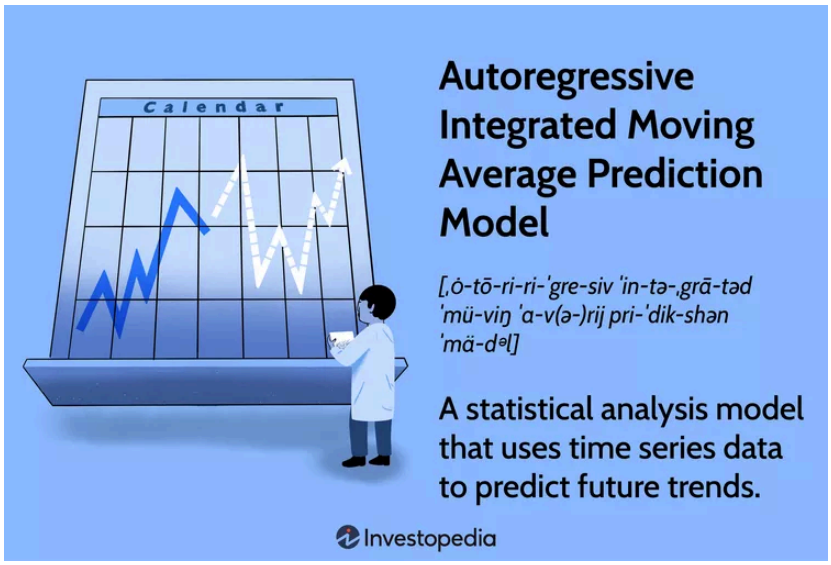


Credit: Investopedia / Dennis Madamba

## What Is an Autoregressive Integrated Moving Average (ARIMA)?

The Autoregressive Integrated Moving Average (A
is a crucial tool in statistical analysis, primarily ut
for forecasting and understanding time series dat
examining the differences between past and futu
values, ARIMA assists in predicting trends in finan
markets, such as stock prices or company earning
based on historical performances. This model
ssive properties, making it

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KEY TAKEAWAYS

- The ARIMA model is a statistical tool used to analyze time series data to understand trends or predict future outcomes, often applied in financial markets.
- ARIMA combines autoregressive and moving average components, represented by the parameters p, d, and q, to capture trends, cycles, and seasonality in data.
- Differencing is applied in ARIMA to make data stationary, ensuring constancy over time and enabling more accurate predictions by removing trends.
- While ARIMA models excel in short-term forecasting, they are less effective for long-term predictions and turning points, owing to their reliance on historical data.
- Building an ARIMA model involves selecting the appropriate order of regression, differencing, and moving averages based on analyzing autocorrelations in the data.

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# How ARIMA Models Analyze Time Series Data

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differences within the series, not the actual values.

An ARIMA model can be understood by outlining each of its components as follows: [1]

- Autoregression (AR): refers to a model that shows a changing variable that regresses on its own lagged, or prior, values.
- **Integrated (I)***:* represents the differencing of raw observations to allow the time series to become stationary (i.e., data values are replaced by the difference between the data values and the previous values).
- Moving average (MA): incorporates the dependency between an observation and a residual error from a moving average model applied to lagged observations.

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## Key Parameters of ARIMA

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as: [1]

- *p*: the number of lag observations in the model, also known as the lag order.
- *d*: the number of times the raw observations are differenced; also known as the degree of differencing.
- *q*: the size of the moving average window, also known as the order of the moving average.

For example, a linear regression model includes the number and type of terms. A zero value for a parameter means that specific component isn't used in the model. This way, the ARIMA model can be constructed to perform the function of an ARMA model, or even simple AR, I, or MA models. [2] [3]

FAST FACT

Because ARIMA models are complicated and work best on very large data sets, computer algorithms and machine learning techniques are used to compute them.

## The Role of Stationarity in ARIMA Modeling

In an ARIMA model, data is differenced to achieve stationarity. A stationary model shows consistent over time. Most economic and market data show trends, so the purpose of differencing is to remov structures. [1]

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computations.

> **IMPORTANT**
>
> A one-time shock will affect subsequent values of an ARIMA model infinitely into the future. Thus, effects of past financial crises can persist in today's autoregressive models.

## Building Your ARIMA Model: A Step-by-Step Guide

1. Download and collect comprehensive price data for the asset you're analyzing.

2. Identify any trends within the data set.

3. Determine the lowest order of differencing (d) by observing the autocorrelations.

4. Check if the series is already differenced by observing if lag-1 autocorrelation is zero or negative.

5. Further difference the series, if necessary, whe
1 is higher than zero.

6. Decide the order of regression (p) and moving average (q) by analyzing autocorrelations and pa
autocorrelations.

7. Use this information to select the appropriate A
[4]

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using past data but require caution. Unlike investment disclaimers, ARIMA assumes past performance affects future values and uses past data for forecasts.

The following table lists other ARIMA traits that demonstrate good and bad characteristics.

| Pros | Cons |
| --- | --- |
| • Good for short-term forecasting<br>• Only needs historical data<br>• Models non-stationary data | • Not built for long-term forecasting<br>• Poor at predicting turning points<br>• Computationally expensive<br>• Parameters are subjective |

## What Is ARIMA Used for?

ARIMA is a method for forecasting or predicting fu
outcomes based on a historical time series. It is b
on the statistical concept of serial correlation, wh
past data points influence future data points.

## What Are the Differences Between Autoregressive and Moving Average Models?

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process is one in which the current value is based on the previous two values. A moving average is a calculation used to analyze data points by creating a series of averages of different subsets of the full data set to smooth out the influence of outliers. As a result of this combination of techniques, ARIMA models can take into account trends, cycles, seasonality, and other non-static types of data when making forecasts.

### How Does ARIMA Forecasting Work?

ARIMA forecasting is achieved by plugging in time series data for the variable of interest. Statistical software will identify the appropriate number of lags or amount of differencing to be applied to the data and check for stationarity. It will then output the results, which are often interpreted similarly to that of a multiple linear regression model.

## The Bottom Line

The ARIMA model is used as a forecasting tool to predict how something will act in the future based on past performance. It is used in technical analysis to predict an asset's future performance.

ARIMA is generally not suitable for forecasts beyo
months as it relies on past data and subjective
parameters. For this reason, it is best used with o
technical analysis tools to get a clearer picture of
asset's performance.

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